

Model Selection, Evaluation, and Tuning

1 Objective

The purpose of this lab is to explore model selection, evaluation, and hyperparameter tuning using two widely-used real-world datasets. Students will:

- Compare the performance of a naive train-test split with k -fold cross-validation.
- Implement hyperparameter tuning using `GridSearchCV` (or `RandomizedSearchCV`) for Logistic Regression (binary classification) and Linear Regression (regression).
- Compare evaluation metrics: Accuracy and F1 Score for classification; Mean Squared Error (MSE) and R^2 for regression.
- Display visualizations directly in the document for all comparisons.

2 Datasets and Preprocessing

2.1 Dataset 1: Breast Cancer Wisconsin (Diagnostic) Dataset

- **Task:** Binary classification using Logistic Regression.
- **Preprocessing Steps:**
 - a. Handle missing values if present.
 - b. Split the dataset into features (X) and binary target labels (y).
 - c. Normalize or standardize the features.
- **Code Hint:** Load the dataset using `sklearn.datasets.load_breast_cancer()`.

Listing 1: Load Breast Cancer Dataset

```
from sklearn.datasets import load_breast_cancer
data = load_breast_cancer()
X_class = data.data
y_class = data.target
```

2.2 Dataset 2: California Housing Dataset

- **Task:** Regression using Linear Regression.
- **Preprocessing Steps:**
 - a. Handle missing values if present.
 - b. Split the dataset into features (X) and the continuous target variable (y).
 - c. Apply feature scaling as needed.
- **Code Hint:** Load the dataset using `sklearn.datasets.fetch_california_housing()`.

Listing 2: Load California Housing Dataset

```
from sklearn.datasets import fetch_california_housing
data = fetch_california_housing()
X_reg = data.data
y_reg = data.target
```

3 Experiment Tasks

3.1 Task 1: Evaluation with Train-Test Split vs. Cross-Validation

- a. **Naive Train-Test Split:**
 - Split each dataset into a training set (e.g., 70%) and a testing set (e.g., 30%).
 - Train the model:
 - For the Breast Cancer Dataset: Use Logistic Regression.
 - For the California Housing Dataset: Use Linear Regression.
 - Evaluate the models on the testing set using:
 - For classification: Accuracy and F1 Score.
 - For regression: Mean Squared Error (MSE) and R^2 .
- b. **k -Fold Cross-Validation:**
 - Apply k -fold cross-validation (suggest $k = 5$ or $k = 10$) using `cross_val_score`.
 - Report the mean and standard deviation of the evaluation metrics.
- c. **Image Display Requirement:**
 - Display a visualization comparing the evaluation metrics obtained from the naive train-test split and k -fold cross-validation.
 - The image must clearly label the metrics (Accuracy, F1 Score, MSE, R^2) and highlight performance differences.

3.2 Task 2: Hyperparameter Tuning with GridSearchCV (or RandomizedSearchCV)

a. Logistic Regression (Breast Cancer Dataset):

- Define a parameter grid with the following suggested hyperparameters:
 - `C` (inverse regularization strength): $\{0.01, 0.1, 1, 10, 100\}$
 - `penalty`: $\{'l1', 'l2'\}$
 - Solver options (choose according to the selected penalty, e.g., `liblinear` or `saga`)

b. Linear Regression (California Housing Dataset):

- Compare plain Linear Regression with regularized versions (Ridge or Lasso Regression). For these models, consider tuning:
 - Regularization strength α : $\{0.001, 0.01, 0.1, 1, 10\}$

c. Implementation:

- Use `GridSearchCV` (or `RandomizedSearchCV`) with cross-validation to perform hyperparameter tuning.
- Report the best hyperparameters along with the corresponding evaluation metrics.

d. Image Display Requirement:

- Display a visualization (e.g., a heatmap or line chart) that shows the performance across different hyperparameter combinations.
- Clearly annotate the plot with the best hyperparameter setting and its corresponding performance.

3.3 Task 3: Comparison and Analysis

a. Evaluation Metrics:

- **Classification (Breast Cancer Dataset):** Accuracy and F1 Score.
- **Regression (California Housing Dataset):** Mean Squared Error (MSE) and R^2 .

b. Comparative Visualization Requirement:

- Display visualizations that compare the evaluation metrics between the naive train-test split and k -fold cross-validation.
- Visualizations should clearly highlight the differences in performance (e.g., using side-by-side bar plots or overlaid line charts).

c. Analysis:

- Discuss how different data splitting strategies affect the evaluation metrics and model stability.

Submission Requirement:

1. Complete in a Jupyter notebook file, name your file as “name_student number#.ipynb” and submit it through “Lab 7 Programming Assignment” on iSpace.
2. Please provide necessary outputs to support your assignment. (e.g., printing necessary information or visualizing through graphs and tables).

Deadline: 11:59pm, April.10, 2026. Late submission = 0; Plagiarism (cheating) = F.